

Optimizing Chemotherapy Safety: Implementing DPYD Pharmacogenomic Testing for 5-Fluorouracil Dosing

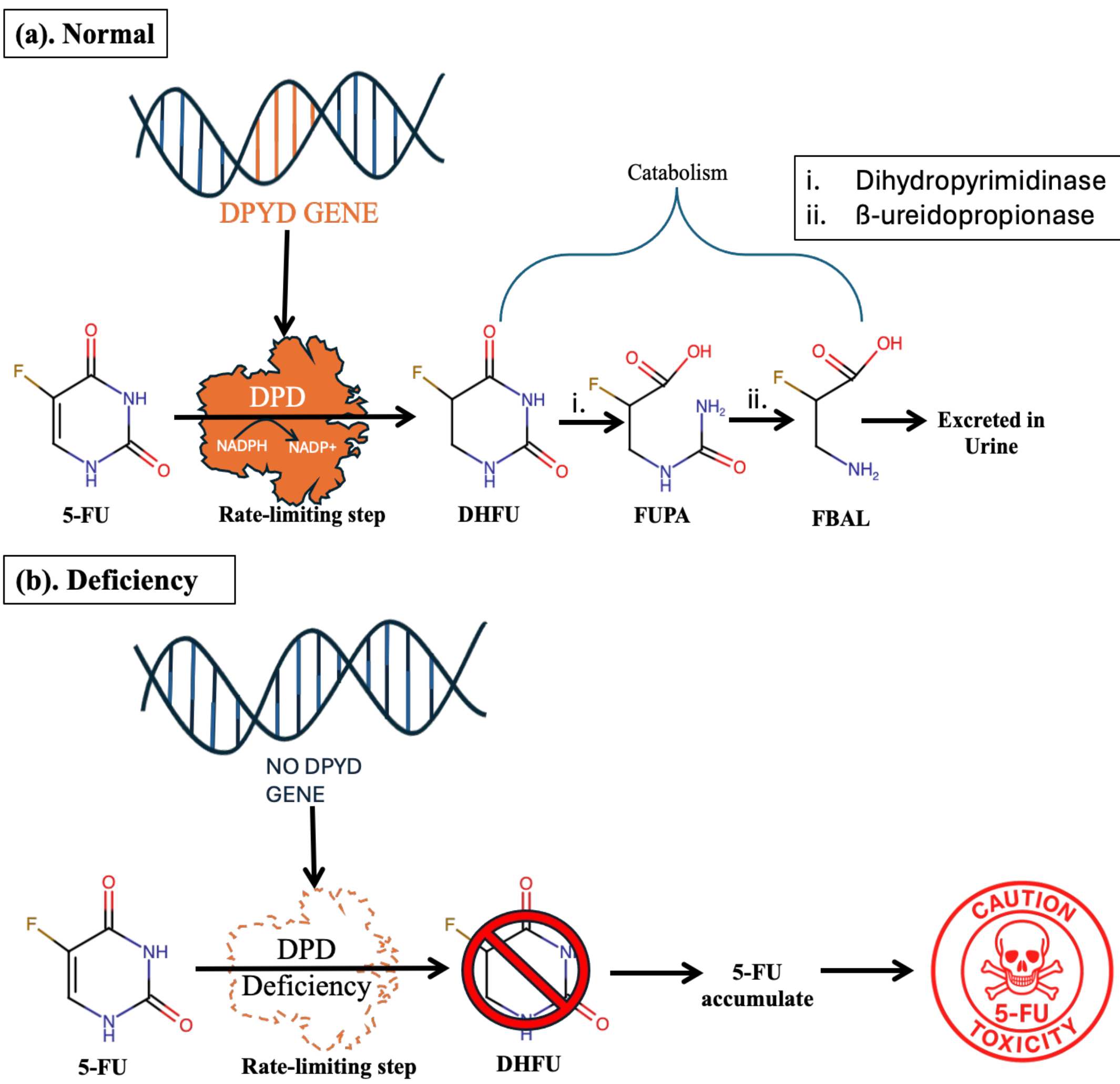
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Background

5-Fluorouracil (5-FU) is a chemotherapy drug used to treat various solid tumors. Its metabolism depends on the enzyme dihydropyrimidine dehydrogenase (DPD), encoded by the *DPYD* gene. Genetic variants in *DPYD* can reduce enzyme activity, leading to toxic buildup of 5-FU and serious side effects. Pharmacogenomic testing identifies patients with these variants, allowing for safer, personalized dosing. The FDA now recommends *DPYD* genotyping prior to initiating 5-FU therapy.

DPYD Gene Function and Deficiency Consequences



Objective and Methods

To evaluate the feasibility and clinical impact of implementing *DPYD* genotyping in oncology settings for safer and more personalized 5-FU chemotherapy dosing.

Methods:

- We conducted a focused review of online resources and peer-reviewed literature to answer key questions related to safety, implementation, and outcomes of *DPYD* testing. The process followed these steps:
- Searched PubMed for prior studies on *DPYD* gene testing and 5-FU toxicity.
 - Selected the most relevant, unbiased, and reliable articles to address clinical impact and feasibility.
 - Used the CPIC website to identify current dosing recommendations based on *DPYD* genotype.
 - Consulted PharmGKB to evaluate the strength and clinical relevance of pharmacogenomic evidence.
 - Reviewed multiple testing sites and external sources to gather ranges for:
 - Costs of DPYD Testing
 - Costs of Treatment (Based on Cancer Type, Number of Cycles, and Regimen Variability)
 - Costs of Treating Toxicities (Per event)
 - Performed a Cost Effectiveness Analysis using these ranges, acknowledging variability in testing and treatment costs.

Results

5- FU Treatment Related Mortality, by Genotype

DPYD Variant	Patients Tested (n)	Variant Carriers (n, %)	Deaths in Variant Carriers (n)	Risk of Death in Variant Carriers (%)
c.1129-5923C>G	6,242	241 (3.9%)	1	0.4 (0.1-2.9)
c.1679T>G	8,799	17 (0.2%)	1	5.9 (0.8-32)
c.1905+1G>A	13,929	183 (1.3%)	8	4.4 (2.2-8.5)
c.2846A>T	10,759	127 (1.2%)	5	3.9 (1.7-9.1)

DPYD Dosing Recommendations (CPIC Guidelines)

Metabolizer Status	Activity Score	Dosing Recommendation	Evidence Level
Normal Metabolizer	2.0	Use standard 5-FU dose.	Strong
Intermediate Metabolizer	1.0 – 1.5	Reduce starting dose by 50%. Titrate based on tolerance and clinical response.	Activity score 1: Strong Activity score 1.5: Moderate
Poor Metabolizer	≤ 0.5	Avoid 5-FU if possible. If used, drastically reduce the dose and monitor closely.	Strong

Cost-Effectiveness of DPYD Testing Prior to 5-Fluorouracil Treatment

Parameter	No DPYD Testing	DPYD Testing
Testing Cost	\$0	\$150-\$300 (Avg \$250)
Treatment Cost (per cycle)	\$10,000-\$30,000 (avg \$20,000)	\$10,000-\$30,000 (avg \$20,000)
Toxicity Treatment Cost (per event)	\$1,000-\$10,000 (avg \$5,000)	\$1000-\$10,000 (avg \$5,000)
Average Total Cost per Patient	\$21,500	\$ 20,500
Prevalence in the Population	3-8%	-
Mortality Rate from Toxicity	0.5-1%	0.1%
Severe Toxicity Rate	30%	<10%
Expected QALYs (No Toxicity)	0.63	0.86
Expected QALYs (Severe Toxicity)	0.03	0.05
Total QALYs per Patient	0.66	0.91
Δ Cost (Testing – No Testing)	-\$1,000	-
Δ QALYs (Testing – No Testing)	+0.25	-
ICER (Cost per QALY Gained)	-\$4,000 (Testing is Dominant)	-

Conclusion

Implications:

Pharmacogenomic testing for *DPYD* variants before initiating 5-FU therapy enables personalized chemotherapy dosing and significantly reduces severe toxicities. Integrating this testing into clinical workflows through EHR decision support improves patient safety and provider adherence.

Future Directions:

- Expand testing to include capecitabine and uracil derivatives.
- Improve clinician education around PGx result interpretation.
- Study real-world cost-effectiveness across broader populations

Ideal Patient Case

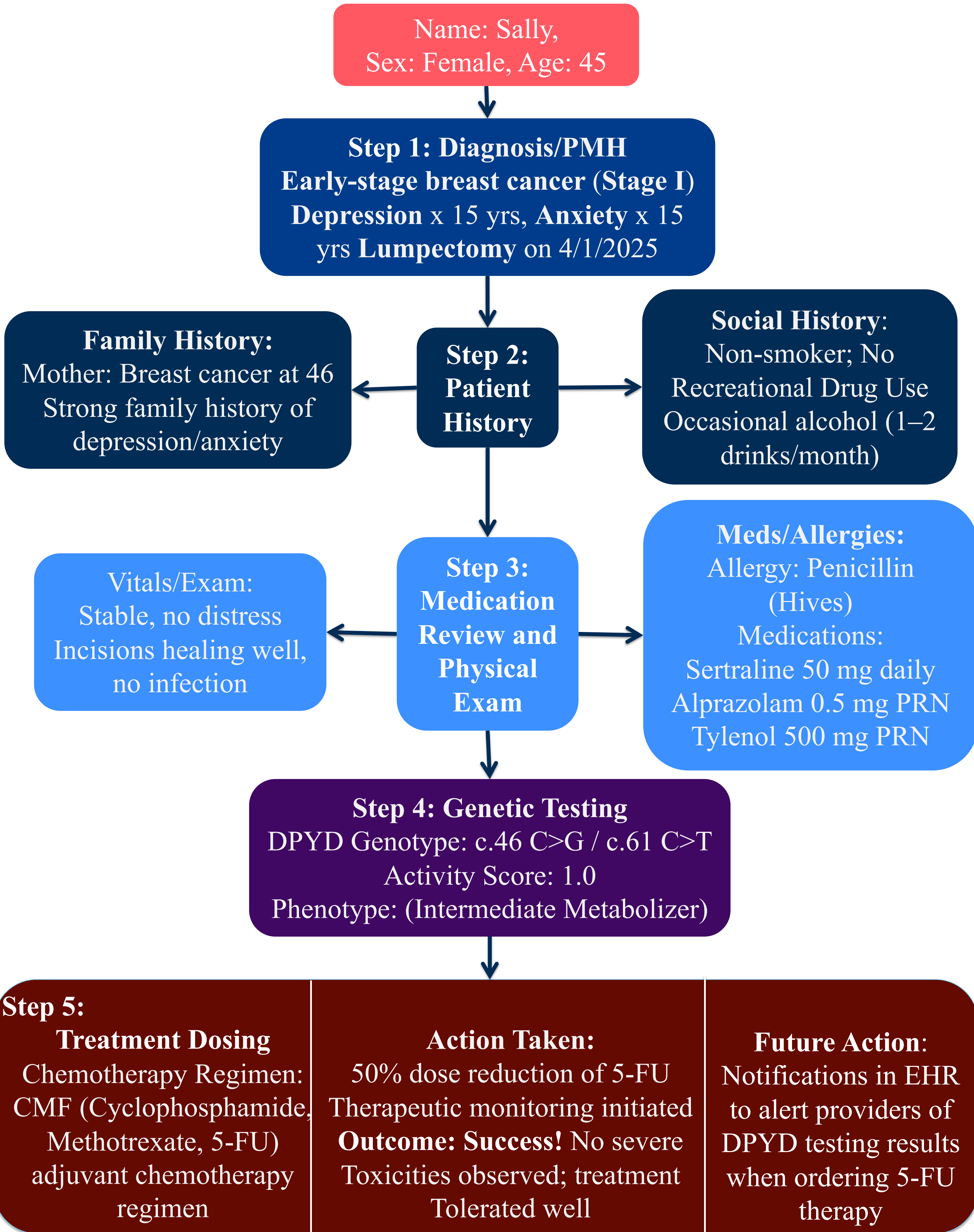


Figure: This flowchart illustrates the key steps in Sally’s treatment journey, leading to a safe and successful chemotherapy outcome.

Acknowledgements

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References

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Patient Education

